

Fix the *System.*

A Maintenance Manager's Field Guide to Getting Work Under Control in 90 Days.

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FIRST EDITION · APRIL 2026

Why This Playbook Exists

> *Monday morning at a mid-size plant. The operations review runs from 8 to 8:45. Near the end, the plant manager asks about the three unplanned outages on Line 4. The maintenance manager explains. The gearbox that failed was the same one that failed in January. And in August. And last October.*

The plant manager nods. Says something about "getting to the root cause." The meeting ends. Nothing changes. The gearbox will fail again in eight to ten weeks.

This is the shape of the problem. Not one person doing anything wrong. Not one moment of negligence. Just a plant that has quietly accepted that some things fail every few months and nobody is going to stop them. Until one of those things stops the plant for a week. Or worse.

If you are a maintenance manager at a mid-size plant, this scene is familiar. You already know what's broken. The PMs are slipping. The backlog is growing. A specific pump on a specific line has its own failure calendar. You do not need another diagram of the reliability pyramid. You need a way to get the next thing fixed before it becomes the thing that shows up on Monday morning for the fourth time.

This playbook gives you that.

WHAT THIS IS

A field guide for the maintenance manager who sees the problems but cannot always get the budget to fix them. Built on SMRP best practices, ISO 55000, Doc Palmer's planning discipline, and RCM fundamentals. Written in concrete steps, not frameworks about frameworks.

WHAT THIS ISN'T

A software pitch. A book about "the future of maintenance." A list of talking points to justify hiring a consultant.

HOW TO USE IT

Twenty-five pages. Four parts.

Part 2 — Foundations. The work management cycle: identify, plan, schedule, execute, close out. No KPI means anything if the underlying cycle is broken. Start here if you are not sure where to start.

Part 3 — Where Are You. A self-assessment. Maturity model and benchmark snapshot. Tells you honestly where you sit, not where last year's audit said you sat.

Part 4 — The Ten Playbooks. Ten specific plays for ten specific KPI failures. Each has a ninety-day roadmap. Pick the one that matches your worst number right now.

Part 5 — Tools and Cadences. RCA quick guide, a ninety-day implementation sequence, the full reference list.

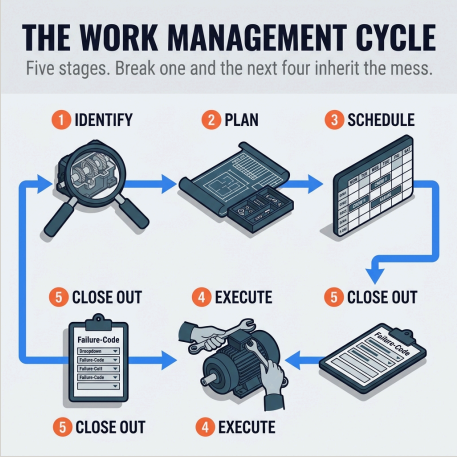
ONE THING BEFORE WE START

None of this matters if the line supervisor doesn't enforce the standard. No CMMS, no playbook, no benchmark fixes a plant where the people running the work don't believe in it. Fix the system. Then back the people running it. That is the job.

Expectations high. Excuses low.
Let's get into it.

Why Foundations Matter

Most plants try to fix their maintenance program by buying software, hiring consultants, or rolling out a new KPI dashboard. None of those things work if the underlying work management cycle is broken.



The cycle is not complicated. Five stages:

1. **Identify** — something needs to be done, and a record of it gets created.
2. **Plan** — the work gets made ready to execute. Parts, procedure, hours, safety.
3. **Schedule** — the work gets committed to a specific day with a specific crew.
4. **Execute** — the work gets done. Safely. With discipline. With the data captured.
5. **Close out and learn** — the record reflects what actually happened, and the system uses the information.

Most plants have something happening at every stage. Very few have all five stages working together. When one stage breaks, the next four inherit the mess.

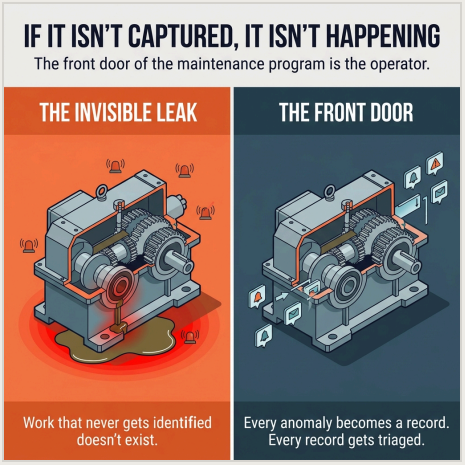
A planning office that plans beautifully cannot recover from a scheduling function that ignores parts availability. A tight schedule cannot survive an execution culture that closes every WO with "Repaired. RTO."

What follows is five pages. One per stage. Each page gives you what good looks like, the most common failure patterns, a cadence for running it, and where to read more.

This part of the playbook is not glamorous. It is the plumbing. But everything in Parts 3, 4, and 5 assumes the plumbing works.

Work Identification

> *Tuesday morning at a mid-size plant. An operator sees a small oil leak under a gearbox. He grabs a rag, wipes it down, doesn't mention it. Shift changes. The next operator sees the same leak, assumes it was already reported. Six weeks later the gearbox seizes mid-shift and takes out two days of production.*



Nobody did anything wrong. The work just never got identified.

Identification is the front door of the maintenance program. If work doesn't get captured, nothing downstream matters. Most plants over-focus on their PM program and under-focus on identification. PMs catch what's scheduled. Everything else has to be captured by people who weren't hired to capture it.

WHAT GOOD LOOKS LIKE

Operators write notifications. Not maintenance. Not the supervisor. The person closest to the equipment. Every notification gets triaged within 24 hours into three buckets: immediate action, planning queue, or reject with reason. Notifications have enough detail to be actionable. Not "pump broken." "Pump 43 discharge pressure fluctuating between 40 and 60 PSI during morning startup. Bearing whine from the coupling end."

COMMON FAILURES

The "I'll tell Mike" culture. Operators report verbally to their favorite tech. Mike forgets. The notification never exists. Notifications go to a queue nobody owns. The initiation form takes 15 minutes to fill out — that's not identification, that's a filter.

CADENCE

Daily: Coordinator triages the new queue. Zero-inbox rule. **Weekly:** Ops and maintenance review the reject pile together. **Monthly:** Pull the data. Low numbers usually mean the channel is broken.

FURTHER READING

- SMRP Best Practice 5.3.3 — Work Order Generation and Identification
- Terry Wireman, *Benchmarking Best Practices for Maintenance* (work identification chapter)
- ISO 14224:2016 — Collection and exchange of reliability and maintenance data

Planning

> *Every plant has a Senior Maintenance Planner. His name is on the org chart. His desk has two monitors and a printer that always needs toner. Everybody likes him. When a breakdown happens at 6 AM he is the first call. When the shutdown gets pushed he is the one who renegotiates with operations. Last month, across four weeks, he planned three jobs. The other six hundred went to schedule without him.*

He is not a bad planner. He is a planner who is never allowed to plan.

A real planner does one thing. They take a piece of work that got identified and make it ready to execute. Parts on hand. Procedure written. Hours estimated within reason. Safety reviewed. When a job leaves the planner's desk, a technician should be able to walk to the asset and start working.

The industry data is consistent. A well-planned job takes about one-third the labor hours of the same job done unplanned. Doc Palmer's research shows one real planner can support six to thirty technicians. If your planner is working on today's breakdown, they are not planning. They are firefighting in a different chair.

WHAT GOOD LOOKS LIKE

- **80% of work is planned before it's scheduled.** Not a goal. A gate.
- **Every planned job has a job plan.** Steps, parts list, hour estimate, special tools, safety requirements, drawings where needed.
- **Parts are kitted and staged 48 hours before execution.** This single practice kills the number-one cause of schedule breakdown.
- **Hour estimates land within 20% of actual** on routine work. Off by 50% every time isn't an estimate. It's a wish.
- **Planners plan. They do not expedite.** Parts chasing is a different job.

COMMON FAILURES

- The planner is actually a clerk. Opens the WO, maybe pulls parts, never builds a real plan.
- Jobs released to schedule without parts. Every week. Nobody flags it.
- Job plans are three words in the description field.
- The planner gets dragged into every breakdown because he knows the assets best. The planning queue never gets touched.

THE READY-TO-SCHEDULE GATE

Before a WO can be scheduled, it must have: (1) parts confirmed *available*, not ordered; (2) job plan attached; (3) hour estimate and craft; (4) asset access window confirmed; (5) safety requirements documented. If any is missing, the WO stays in planning. Held consistently, this one rule moves schedule compliance up twenty points inside 90 days.

CADENCE

Daily: 30 min, first thing — planner processes notification queue. **Weekly:** Planning meeting with supervisor and parts lead. **Monthly:** Audit planned vs actual hours. Review job plan quality.

FURTHER READING

- Doc Palmer, *Maintenance Planning and Scheduling Handbook* (3rd ed.) — the bible
- SMRP Best Practice 5.3.2 — Work Order Planning
- ISO 55001:2014 Clause 8.1 — Operational planning and control

Scheduling

> *The Friday scheduling meeting runs forty minutes over every week. Maintenance reads off next week's schedule. Operations agrees to everything. Monday morning at 7 AM, production calls about a priority changeover on Line 3 that nobody mentioned Friday. Two planned jobs get bumped. By Tuesday afternoon the schedule is a fiction nobody references.*



This is not a scheduling problem. It is a commitment problem. A real schedule is a contract. It says: we will make the equipment available during these windows, you will have the right crews with the right parts to do the right work. Both sides sign up. Breaking the contract should require a conversation, not a text message at 7 AM.

The industry target is 80 percent-plus schedule compliance. Most plants run 40 to 60. The gap is not craft. It is discipline.

WHAT GOOD LOOKS LIKE

The schedule is locked 48 to 72 hours before the work week starts. This is the frozen zone. No changes unless safety-critical. The weekly meeting has operations, maintenance, and the storeroom in the same room. If parts are not ready, the WO goes back to planning. A two-week look-ahead is published. One person owns schedule compliance as a number.

COMMON FAILURES

The "emergency" that isn't. A priority changeover someone has known about for three weeks shows up as urgent Monday morning. Schedule built without confirming labor — on paper crew is ten, Monday morning it's five. No frozen zone; the schedule becomes a list of things maintenance hopes to do.

CADENCE

Weekly: Scheduling meeting, 30 min. Lock next week. Review last week's compliance. **Daily:** Ten-minute morning huddle. **Monthly:** Review schedule breaks by category. Fix the top one.

FURTHER READING

- Doc Palmer, *Maintenance Planning and Scheduling Handbook* (3rd ed.) — weekly scheduling discipline
- SMRP Best Practice 5.3.5 — Schedule Compliance
- SMRP Best Practice 5.3.4 — Break-in Work Percentage

Work Execution

> *A senior tech opens up a centrifugal pump scheduled for seal replacement. The seal is shot, as expected. But the coupling is also gone, the bearing housing has scoring, and the shaft has a slight bend. The job plan said two hours. He is now looking at sixteen. He doesn't call planning. Doesn't update the WO. Pulls parts from the storeroom himself, finishes the job at 9 PM. The WO closes Monday with "Repaired. RTO."*

Nothing in that story is exactly wrong. The work got done. And at the same time: the data is worthless, the storeroom is short a coupling nobody logged, and the next time that pump family shows symptoms nobody will know how often the bearing also fails.

Execution is where the maintenance program meets reality. Plans are plans. The floor is the floor. The discipline is not about preventing improvisation. Good techs improvise every day. It is about capturing what actually happened, so the system can learn from it.

WHAT GOOD LOOKS LIKE

- **Safety and LOTO are non-negotiable and pre-staged.** Lockout procedures on the job plan. Locks and tags in the kit. No technician working out LOTO on the fly.
- **Scope changes trigger a notification.** Tech finds more damage? Radio call, planner updates the WO, storeroom pulls more parts.
- **Handoffs between shifts have real content.** Not "pump job, still going." What's done, what's left, what was found.
- **As-found and as-left conditions get documented.** "As-found: coupling worn 80%, shaft TIR 0.012. As-left: new coupling, shaft aligned to 0.002 TIR." This is the data that makes RCA possible six months later.

COMMON FAILURES

- Scope creep that never makes it back to the CMMS. Tech fixes five things while he's in there. The WO documents one.
- LOTO as improv. Procedure not written. Tech figures it out. Mostly works. Until it doesn't.
- The silent handoff. Shift changes mid-job. Next crew walks in cold, spends an hour rediscovering what the last shift already found.
- Closeout notes that say "Repaired. RTO." That is not data. That is a receipt.

CADENCE

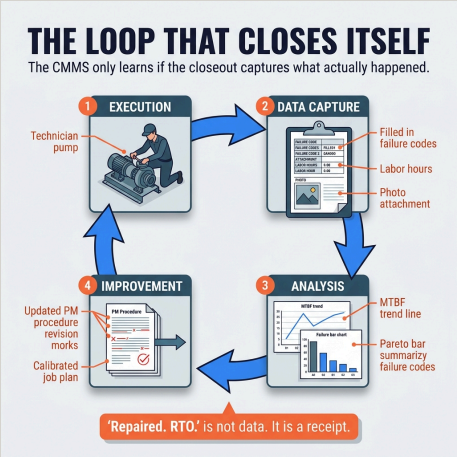
Per job: Safety briefing before starting. Status check at the midpoint on jobs over four hours. Scope change triggers an update same day, not at close. **Daily:** End-of-shift debrief. **Weekly:** Supervisor pulls five random completed WOs and reads them. Anything closed with "RTO" as the only note goes back.

FURTHER READING

- Ron Moore, *Making Common Sense Common Practice* (work execution discipline)
- SMRP Best Practice 5.3.7 — Proactive Work Percentage
- OSHA 1910.147 — The Control of Hazardous Energy (Lockout/Tagout)

Close-Out and the Feedback Loop

> *Six months after the pump job, the same pump fails again. Same symptoms. The reliability engineer pulls the history. Three prior WOs, each closed with variations of "Repaired. RTO." No failure codes. No as-found or as-left notes. No photos. Nothing.*



The equipment is telling you something. The CMMS is not listening.

Close-out is the most neglected part of the cycle because it feels like paperwork. It isn't. Close-out is the difference between a plant that learns and a plant that keeps relearning the same lesson. Everything downstream — RCA, PM optimization, reliability engineering — depends on what gets captured when a WO closes.

Most CMMS implementations have the right fields. What they don't have is enforcement. The fields exist. Techs skip them. Supervisors sign off anyway. Six months of data is garbage that nobody wants to admit is garbage.

WHAT GOOD LOOKS LIKE

Every WO closeout has a failure code, a cause, and an action. Actual labor hours captured and compared to estimate. Photos and notes on anything non-routine. The reliability engineer reviews all WOs on critical assets weekly. A monthly audit pulls 10 random closed WOs and reads them — if more than 2 out of 10 are incoherent, the discipline is broken.

COMMON FAILURES

"Repaired. RTO." as the universal note. Failure codes that don't match the failure — tech picks "Other" every time because the dropdown is 200 options deep. Labor hours that are round numbers only — suspiciously tidy, almost certainly fabricated. No feedback from close-out to planning — the data is captured and ignored. This is the worst failure mode. It creates the illusion of a learning system.

CADENCE

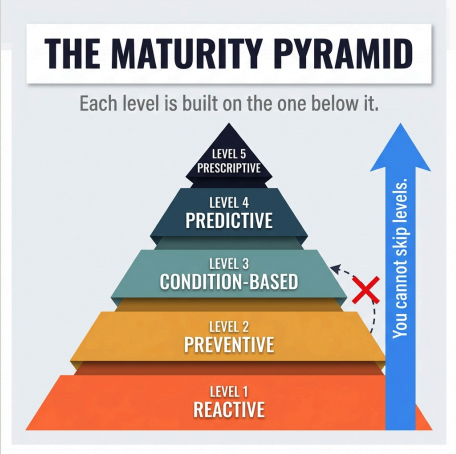
Per job: Closeout at completion, not end of week. **Weekly:** Reliability engineer reviews closed WOs on critical assets. **Monthly:** PM adjustments based on actual labor data. Close-out quality score published on the board, by crew.

FURTHER READING

- John Moubray, *Reliability-Centered Maintenance II* — the feedback loop
- SMRP Best Practice 5.4.1 — Equipment Reliability and Defect Elimination
- ISO 14224:2016 — Failure data collection standards

The Maturity Model

> *Walk into Plant A at 7 AM. The whiteboard is full of work orders stamped URGENT. The planner is on the phone with a vendor. Walk into Plant B at 7 AM. A technician is taking a vibration reading on a critical pump. The planner is reviewing next week's kitted jobs. Same industry. Same equipment. Different maturity level.*



There are five maturity levels in maintenance. Every plant sits at one of them. Most plants don't know which one.

THE FIVE LEVELS

Level 1 — Reactive. Fix on fail. No formal PM program. Reactive work above 50%. Most plants claim they are past this. Many are not.

Level 2 — Preventive. Time-based PM. Compliance 70-90%. Reactive 20-40%. This is where most mid-size plants actually live.

Level 3 — Condition-Based. PM compliance above 90%. Reactive below 20%. Condition monitoring on critical assets. Formal RCA for repeat failures. The plant runs on data.

Level 4 — Predictive. PM compliance above 95%. Reactive below 10%. Predictive models detect failures before they occur.

Level 5 — Prescriptive. AI-driven optimization. Near-zero unplanned downtime. If you are Level 5, you are not reading this playbook.

MOVING UP A LEVEL

You cannot skip levels. A Level 1 plant cannot leapfrog to Level 3 by buying vibration analyzers. Condition monitoring on top of broken PM compliance produces nothing but expensive alerts nobody acts on. Each level builds on the one below it.

The playbooks in Part 4 are tagged by maturity level. Start with the ones that match where you actually are. Get them working. Then move up.

The Self-Assessment

Five questions. Answer honestly. The point is not to feel good. The point is to know.

Q1 — PM COMPLIANCE

What percent of scheduled PMs are completed on time, measured weekly?

Below 50% → Level 1

50-70% → Level 1 or 2

70-90% → Level 2

90-95% → Level 3

Above 95% → Level 4 potential

Q2 — REACTIVE WORK

What percent of WOs are unplanned (reactive, emergency, break-in)?

Above 50% → Level 1

30-50% → Level 1 or 2

20-30% → Level 2

10-20% → Level 3

Below 10% → Level 4 potential

Q3 — BACKLOG MANAGEMENT

No formal backlog → Level 1

Tracked but growing → Level 1 or 2

Tracked, 4-6 weeks, stable → Level 2

Tracked, 2-4 weeks, actively purged → Level 3 or higher

Q4 — CONDITION MONITORING

None → Level 1 or 2

Informal or ad-hoc → Level 2

On most critical assets, integrated to CMMS → Level 3

With trending and predictive models → Level 4

"We send oil samples once a year" is not a program. It is a gesture.

Q5 — ROOT CAUSE ANALYSIS

Repeated same as last time → Level 1 or 2

Supervisor looks at it informally → Level 2

Formal RCA (5 Whys, Fishbone, FMEA) → Level 3

RCAs tracked, closed, corrective actions verified → Level 3+

SCORING

Take the **lowest** level across all five questions. That is where you actually are. Not the average. Maturity is bounded by your weakest stage.

A plant with great PM compliance but no RCA process is not a Level 3 plant. It is a Level 2 plant with an unusually strong PM program.

WHERE TO GO FROM HERE

- **Scored Level 1** → Start with PB-RE-001 and PB-PM-001. These are the foundation plays.
- **Scored Level 2** → Add PB-BK-001 and PB-SC-001. Your PM program exists but your execution discipline doesn't.
- **Scored Level 3** → Focus on PB-FA-001 and PB-CBM-001. You've earned the right to optimize.

The Benchmark Snapshot

Eight KPIs every mid-size plant should track weekly. These are the industry standards. Not aspirations. The numbers that separate a well-run maintenance program from a struggling one.



KPI	UNIT	ACTION	TARGET	WORLD-CLASS
PM Compliance	%	<85	90	≥95
Schedule Compliance	%	<75	85	≥90
On-Time Completion	%	<80	90	≥95
Reactive Work	%	>30	≤20	≤10
Backlog	weeks	>8	4-6	2-3
Inspection Compliance	%	<85	90	≥95
Availability	%	<90	95	≥97
MTTR	hours	>8	<4	<2

Sources: SMRP Best Practices 5.1.1–5.6.1, ISO 55000:2014, SAE JA1012, industry benchmarks.

HOW TO USE THIS TABLE

Pull your own numbers for the last 30 days. Mark each red, yellow, or green. The color pattern tells you where to focus. Three reds is a Level 1 plant pretending to be Level 2. All green is a real Level 3 plant, and there aren't many.

WHERE THE NUMBERS COME FROM

PM Compliance: PMs completed on schedule ÷ PMs due. **Schedule Compliance:** WOs executed on day scheduled ÷ WOs scheduled that day. **Reactive Work:** Unplanned WO labor hours ÷ total WO labor hours. Don't count PMs in the denominator. **Backlog:** Open planned hours ÷ available weekly crew hours. **Availability:** (Total time – unplanned downtime) ÷ total time. **MTTR:** Total repair time on unplanned outages ÷ number of unplanned outages.

A WORD ON MEASUREMENT

If you cannot pull these numbers from your CMMS in under ten minutes, that is a finding in itself. Not a tool problem. A data hygiene problem. Start with PM Compliance and Reactive Work. Get those two working. The others follow.

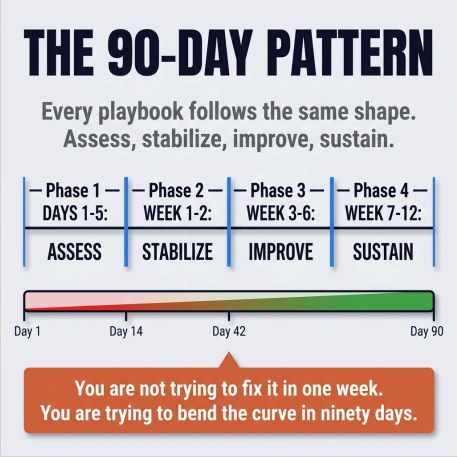
1. PM Compliance Recovery

ID PB-PM-001 **SEVERITY** CRITICAL **LEVEL** 2

> *Trigger: PM compliance below 85% for two or more consecutive periods.*

> *Monday morning. The supervisor opens the PM dashboard. Seventy-three percent. Same as last week. Same as the week before. He sends the number up with a brief note: "working on it." Nothing changes. Two weeks later it is at 68. Then 65.*

PM compliance doesn't collapse all at once. It erodes one missed job at a time. A parts delay here. A break-in there. Every individual decision feels reasonable. The pattern is not.



WHY IT MATTERS

Every 1% drop in PM compliance correlates with a 2-3% rise in corrective work. Plants below 85% run roughly 40% higher maintenance costs than plants above 90. Unexecuted PMs let failure modes develop silently. The assets come back and bite you on a Tuesday afternoon six weeks later.

90-DAY ROADMAP

Days 1-3	Triage the PM backlog. Export overdue and upcoming. Categorize by asset criticality (A/B/C). Critical assets first.
Days 3-5	Root cause the delays. Interview planners and technicians. Top causes: parts, scheduling, labor, procedures.
Week 2	PM kitting and staging. Pre-kit all parts 48 hours before scheduled date.
Week 2-3	Protect PM windows. Frozen zone inside 48 hours. No break-in displaces a due PM unless safety-critical.
Week 3	Automated reminders and escalation (3 days before, day of, +1 day supervisor).
Week 4-6	Review PM frequencies and procedures. Remove tasks that don't address failure modes.
Ongoing	Weekly reporting. Target 90%+ in 8 weeks, 95%+ in 16.

TOP ACTIONS

1. Triage and clear backlog by asset criticality
2. Implement PM kitting 48 hours before execution
3. Install a frozen zone protecting PMs from displacement
4. Set up automated reminder and escalation workflow

BENCHMARK

PM Compliance: world-class ≥95%, target 90%, action threshold <85%. Source: SMRP Best Practice 5.5.6.

FURTHER READING

- SMRP Best Practice 5.5.6 — PM/PdM Work Order Completion Rate
- SAE JA1011 — Evaluation Criteria for RCM Processes
- ISO 55000:2014 — Asset Management System Requirements

2. Repeat Failure Investigation

ID PB-FA-001 **SEVERITY** CRITICAL **LEVEL** 3

> *Trigger: Same asset has 3+ work orders with the same failure code in the period.*

> *A centrifugal pump on Line 3 has failed four times this year. Each time: same symptom, same diagnosis, same repair. Replace the mechanical seal. Return to operation. Close the WO with "Repaired. RTO." On the fifth failure, the reliability engineer finally pulls the history and discovers the asset has eaten seven seals in eighteen months. Nobody had asked why until now.*

Repeat failures are the loudest signal the equipment sends you. They tell you the repair approach is treating symptoms, not causes. The fix-fail-fix cycle consumes two to five times more labor than a permanent fix would cost.

WHY IT MATTERS

Repeat failures are the single strongest predictor of a reliability gap. A plant above 15% repeat failure rate has a reliability culture problem, not a craft problem. The techs are not bad at repairing. The system is bad at asking questions.

90-DAY ROADMAP

Days 1-2	Pull WO history by asset and failure code. Rank by frequency × consequence. Top 3-5 get focus.
Days 3-10	Formal RCA on each. 5 Whys simple, Fishbone multi-factor, FMEA for critical assets.
Week 2	Corrective action plan per root cause. Owner and deadline. PM updates, design mods, training, or CBM.
Week 2-3	Update PM procedures. If failure is detectable via inspection, add or modify PM tasks.
Week 3-6	Install condition monitoring on highest-consequence repeat failures.
Week 6-12	Verify and close. Monitor for 2-3 MTBF intervals. Recurrence escalates to engineering.

TOP ACTIONS

1. Identify top 3-5 repeat combinations by frequency × consequence
2. Run formal RCA (5 Whys, Fishbone, or FMEA)
3. Update PM procedures based on root cause findings
4. Install condition monitoring on highest-consequence assets

BENCHMARK

Repeat Failure Rate: world-class <5%, target <10%, action threshold >15%. Source: SMRP Best Practice 5.4.1.

FURTHER READING

- SMRP Best Practice 5.4.1 — Equipment Reliability
- SAE JA1012 — Guide to the RCM Standard
- MIL-STD-1629A — FMEA military standard

3. Backlog Growth Control

ID PB-BK-001 **SEVERITY** WARNING **LEVEL** 2

> *Trigger: Backlog has been growing for 3+ consecutive periods.*

> *The Friday scheduling meeting. Someone reads out "backlog is now at 9.2 weeks." Nobody responds. Last week it was 8.8. Three months ago it was 6.4. The number has been climbing for twelve straight weeks. It has become wallpaper.*

A healthy backlog is 2 to 4 weeks of planned work. It is a buffer, not a graveyard. When the backlog grows week over week, the plant is creating work faster than it can complete it. That is a leading indicator. It tells you what your reactive work percentage will look like in three months.

WHY IT MATTERS

Above 6 weeks, the plant has lost the ability to respond to emerging priorities effectively. Urgent work can't find a clean schedule slot. Stale WOs pile up. By 8 weeks, routine work takes two months to complete. Technicians stop documenting small defects because the WO will sit there anyway.

90-DAY ROADMAP

Days 1-3	Backlog review and purge. Close or cancel stale WOs (duplicates, fixed, superseded). Typically removes 10-20%.
Days 3-5	Categorize and prioritize. Must-Do (safety/compliance), Should-Do (reliability), Could-Do (nice-to-have).
Week 1	Staffing vs workload. Compare backlog hours to available crew hours. If >6 weeks and growing, contractor support or overtime.
Week 2	Weekly backlog review cadence. 30 min with maintenance leads. Set weekly completion targets.
Week 2-4	Root cause the growth. WO creation climbing? Completion falling? Parts delays? Target the primary driver.

TOP ACTIONS

1. Conduct a backlog purge to remove stale items
2. Categorize and prioritize (Must / Should / Could)
3. Evaluate staffing levels vs current workload
4. Implement a weekly backlog review cadence

BENCHMARK

Backlog: world-class 2-3 weeks, target 4-6, action threshold >8. Source: SMRP Best Practice 5.3.1.

FURTHER READING

- SMRP Best Practice 5.3.1 — Maintenance Backlog
- SMRP Best Practice 5.3.2 — Work Order Completion Rate
- ISO 55001:2014 — Asset Management Requirements

4. Reactive Work Reduction

ID PB-RE-001 **SEVERITY** CRITICAL **LEVEL** 2

> *Trigger: Reactive work exceeds 30% of total work orders.*

> *The shift started with eight planned jobs. At 7:30 AM a gearbox fails on Line 2. Two techs pulled. At 9:15 a conveyor in packaging seizes. Two more pulled. At 11 the plant manager calls about a critical pump warning on the boiler. The shift ends with two planned jobs done, six displaced.*

Some plants run this way for years and call it "being responsive." What it actually is: an expensive way to lose money. Reactive costs three to five times more than planned work. And the displacement cascades — the planned jobs that got bumped become next week's breakdowns.

WHY IT MATTERS

Plants above 30% reactive work operate at 40-60% lower efficiency than plants below 20. Every hour spent reacting is an hour not spent preventing. The loop cannot be broken by working harder. It breaks by working differently.

90-DAY ROADMAP

Days 1-3	Pareto the top 10 failure modes driving unplanned work.
Week 1-2	Map failure modes to PM coverage. No PM? Create one. PM failing to prevent? Revise task or frequency.
Week 2-4	Operator-driven reliability (ODR). Train operators on inspection, cleaning, lubrication, anomaly reporting.
Week 3	Review emergency priority classification. Most plants over-classify routine work as urgent. Tighten criteria.
Week 4-8	Install condition monitoring on high-value assets with repeat failures.
Ongoing	Weekly planned vs reactive ratio reporting. Target 5% reduction per month until below 20.

TOP ACTIONS

1. Pareto the top 10 failure modes driving unplanned work
2. Map failure modes to PM coverage and close gaps
3. Install condition monitoring on critical equipment
4. Implement operator care and anomaly reporting

BENCHMARK

Reactive Work: world-class ≤10%, target ≤20%, action threshold >30%. Source: SMRP Best Practice 5.1.2.

FURTHER READING

- SMRP Best Practice 5.1.2 — Reactive Maintenance Work
- SAE JA1011 — RCM Evaluation Criteria
- Ron Moore, *Making Common Sense Common Practice*

5. Schedule Compliance Improvement

ID PB-SC-001 **SEVERITY** WARNING **LEVEL** 2

> *Trigger: Schedule compliance below 75%.*

> *The weekly scheduling meeting locks next week at 60 hours. Monday, production asks to move a changeover up two days. Tuesday, two techs call out. Wednesday, operations swaps Line 3 priorities. Thursday, half the kitted parts didn't arrive. Friday, 32 of 60 hours got executed. Everyone shrugs and rebuilds for next week, same way, same outcome.*

Schedule compliance below 75% is not a scheduling problem. It is an execution discipline problem. The schedule exists. Nobody is defending it.

WHY IT MATTERS

Below 75% compliance, plants run 30-50% lower labor productivity. Techs context-switch, wait for parts, replan. Schedule breaks drive backlog growth, which drives reactive work, which drives more schedule breaks. The loop feeds itself.

90-DAY ROADMAP

Days 1-3	Analyze break-in patterns. Pareto by cause: failure, management request, parts, absenteeism.
Week 1-2	Frozen schedule window. Lock 48-72 hours before work week. No changes inside zone unless safety-critical.
Week 2-3	Parts and material planning. Kit and stage before work day. No parts, no schedule.
Week 2	Strengthen weekly scheduling meeting. Ops, maintenance, storeroom same room. Lock the week.
Week 3-4	Tighten emergency criteria. Gatekeeper role. Only supervisor authorizes breaks.

TOP ACTIONS

1. Pareto break-in causes and implement prevention
2. Install 48-72 hour frozen schedule window
3. Enforce parts-available rule before scheduling
4. Implement gatekeeper for emergency reclassifications

BENCHMARK

Schedule Compliance: world-class ≥90%, target 85%, action threshold <75%. Source: SMRP Best Practice 5.3.5.

FURTHER READING

- Doc Palmer, *Maintenance Planning and Scheduling Handbook* (3rd ed.)
- SMRP Best Practice 5.3.5 — Schedule Compliance
- SMRP Best Practice 5.3.4 — Break-in Work Percentage

6. Inspection Compliance Recovery

ID PB-INS-001 **SEVERITY** WARNING **LEVEL** 2

> *Trigger: Inspection overdue rate above 15%.*

> *The external inspector shows up Tuesday for the quarterly audit. The plant manager sits in, confident. Twenty minutes in, the inspector asks to see the pressure vessel inspection records. The reliability engineer pulls them up. The last API 510 inspection was 62 days past due. The one before that was 45. The inspector notes it in the report. The room gets very quiet.*

Inspections are the earliest detection opportunity. Each missed inspection extends the window where a developing failure can go undetected. For regulatory inspections, non-compliance is not an internal problem. It is a legal exposure problem.

WHY IT MATTERS

Missed inspections don't just add risk. They compound it. A small corrosion defect missed at month 6 becomes a large defect at month 9 becomes a through-wall leak at month 12. The inspection cycle exists because nobody can see what is happening inside a pressure vessel without looking.

90-DAY ROADMAP

Days 1-3	Prioritize overdue inspections. Regulatory > safety-critical > condition-based. Clear regulatory first.
Days 3-5	Identify bottlenecks. Conflicts with PMs? Inspector capacity? Narrow access windows?
Week 2-3	Mobile inspection tools. Tablet checklists reduce completion time 30-50%. Real-time compliance tracking.
Week 2	Automated reminders and escalation. 3 days before, day of, +1 supervisor, +2 manager.
Week 3-4	Review inspection frequencies. Some too frequent (waste). Others too infrequent (risk).

TOP ACTIONS

1. Prioritize regulatory and safety-critical inspections first
2. Implement mobile inspection tools
3. Install automated reminders and escalation workflow
4. Audit inspection frequencies against actual condition data

BENCHMARK

Inspection Compliance: world-class ≥95%, target 90%, action threshold <85%. Source: SMRP Best Practice 5.5.6.

FURTHER READING

- API 510 / 570 / 653 — Pressure Vessel / Piping / Tank Inspection Codes
- SMRP Best Practice 5.5.6 — PM/PdM Work Order Completion
- ISO 55000:2014 — Asset Management System Requirements

7. Spare Parts Management

ID PB-SPM-001 **SEVERITY** WARNING **LEVEL** 3

> *Trigger: Average WO completion time exceeds 2× estimate, with 3+ completed WOs affected.*

> *The storeroom has 42,000 line items. The maintenance manager can recite this fact. What he cannot do is explain why, on a given Tuesday, a technician is waiting three days for a specific bearing on a specific pump. The bearing exists in the inventory system. Nobody knows if any are actually in the bin until someone walks over and looks.*

Thirty to fifty percent of maintenance delays are parts-related. It is the single most chronic cause of MTTR inflation. And it is not usually about not spending enough on inventory. Most plants have the wrong inventory, kept badly.

WHY IT MATTERS

Emergency parts procurement costs three to ten times normal price. Waiting for parts extends MTTR by 50-200%. A stock-out on a critical spare, the one bearing you needed that isn't there, can turn a breakdown that should have taken hours into one that takes days.

90-DAY ROADMAP

Days 1-5	Analyze parts-related WO delays. Review WOs where actual exceeded estimate. Interview techs.
Week 1-2	ABC classification. A-items (critical, high consequence), B-items (important, substitutable), C-items (low-cost).
Week 2-3	Set reorder points for critical spares. ROP = (avg daily usage × lead time) + safety stock.
Week 3-4	PM parts kitting. BOM for scheduled PMs. Kit and stage 48 hours before date.
Week 4-6	Vendor partnerships for fast-turnaround items. Consignment stock or emergency delivery agreements.

TOP ACTIONS

1. Analyze WO delays for parts-related bottlenecks
2. Implement ABC classification for spare parts
3. Set reorder points and safety stock for critical spares
4. Create a PM parts kitting process

BENCHMARK

Parts-Related Delays: world-class <3%, target <5%, action threshold >10%. Stockout Rate: world-class <2%, target <5%, action >10%. Source: SMRP Best Practice 5.6.1.

FURTHER READING

- SMRP Best Practice 5.6.1 — MRO Inventory Management
- ISO 55001:2014 — Asset Management Requirements
- Terry Wireman, *Benchmarking Best Practices* — inventory management chapter

8. Condition-Based Monitoring

ID PB-CBM-001 **SEVERITY** INFO **LEVEL** 3

Trigger: MTBF < 500h on critical assets with repeat failures.

> *A critical centrifugal compressor fails approximately every 400 operating hours. Same failure mode every time: bearing degradation, then seal failure. The PM program says inspect the bearings quarterly. Quarterly inspections are taking quarterly photographs of a process that completes in three months. By the time the next PM comes around, the bearing is already going. The plant is catching the failure, not preventing it.*

Time-based PMs cannot prevent failures whose timing is driven by operating condition, not calendar. That is what CBM exists for. Vibration data would show that bearing degrading two weeks before failure. Oil analysis would show metal particles climbing. The data is there. The plant is not collecting it.

WHY IT MATTERS

CBM reduces unplanned downtime by 30-50% and extends asset life by 20-40% versus time-based PM. It is the key transition from Level 2 to Level 3. Mature CBM programs achieve MTBF improvements of 200-500% on monitored assets.

90-DAY ROADMAP

Days 1-5	Identify CBM candidates. High criticality, MTBF <500h, repeat failures. Pick 3-5 for pilot.
Week 1-2	Select monitoring techniques. Vibration (rotating), thermography (electrical), oil analysis (lubricated), ultrasound (air/bearings).
Week 2-3	Establish baseline readings. Take initial measurements on healthy assets.
Week 3-4	Define alert and alarm thresholds per ISO 10816 (vibration), NETA (thermography).
Week 4-6	Integrate with CMMS workflow. Alerts auto-generate WOs with condition data attached.
Week 4-8	Build internal expertise. Train 1-2 techs. Certifications for long-term capability.

TOP ACTIONS

1. Identify 3-5 critical assets with low MTBF as CBM pilots
2. Select appropriate monitoring technique per failure mode
3. Establish baseline readings and alert thresholds
4. Train technicians on data collection and basic analysis

BENCHMARK

MTBF (monitored): world-class >2000h, target >1000h, action threshold <500h. PdM Coverage: world-class >80%, target >50%, action <20%. Source: SAE JA1012, SMRP 5.5.2.

FURTHER READING

- SAE JA1012 — Guide to the RCM Standard
- ISO 17359:2018 — Condition Monitoring and Diagnostics
- ISO 10816 — Mechanical Vibration Evaluation
- SMRP Best Practice 5.5.2 — PdM Program Coverage

9. Downtime Reduction

ID PB-DT-001 **SEVERITY** CRITICAL **LEVEL** 2

> *Trigger: Availability below 95%, or unplanned downtime exceeds target.*

> *The morning production meeting. The production manager reads off last week's numbers. Availability on Line 4: 91 percent. Availability on Line 2: 93. The numbers have been in the low 90s for six months. Maintenance says "we're working on it." Production says "okay." Nobody asks what "working on it" actually means.*

Every 1% drop in availability translates to significant lost production capacity. World-class plants target >97%. Plants below 95 typically have 30-50% higher maintenance costs. Unplanned downtime costs three to ten times more than planned because of emergency response, overtime, and expedited parts.

WHY IT MATTERS

Availability is the KPI that executives understand. It maps directly to throughput, to revenue, to the numbers on the monthly report. When availability is low, everything else in maintenance lives downstream of that conversation. Get availability above 95 and your relationship with operations changes.

90-DAY ROADMAP

Days 1-2	Pull downtime data by asset for 90 days. Rank by hours. Top 5-10 assets typically 60-80% of total.
Days 2-4	Break down by reason category. Unplanned, planned, material, operator, setup, quality. Identify dominant.
Week 1-2	RCA on top 3-5 downtime-contributing assets. 5 Whys or Fishbone.
Week 2-3	Improve PM coverage for high-downtime assets. Add or modify tasks. Align frequencies with MTBF.
Week 3-4	Reduce MTTR through preparation. Pre-stage spares. Standardize job plans. Tools and manuals accessible.
Ongoing	Weekly monitoring by asset group and shift. Target 1-2% availability improvement per month to reach 97%.

TOP ACTIONS

1. Pareto the top 5 downtime-contributing assets
2. Conduct RCA on worst offenders
3. Review PM procedures for high-downtime equipment
4. Install condition monitoring on critical assets without CBM

BENCHMARK

Availability: world-class ≥97%, target 95%, action threshold <90%. Unplanned Downtime: world-class <3%, target <5%, action >10%. Source: SMRP Best Practice 5.4.2.

FURTHER READING

- SMRP Best Practice 5.4.2 — Overall Equipment Effectiveness
- ISO 55000:2014 — Asset Management System Requirements
- SAE JA1012 — Guide to the RCM Standard

10. Asset Reliability Improvement

ID PB-DT-002 **SEVERITY** WARNING **LEVEL** 3

> *Trigger: MTTR exceeds 4 hours, or critical assets have recurring downtime events.*

> *A critical motor fails at 2 AM. The night tech diagnoses the failure in fifteen minutes (bearing). He then spends the next four hours trying to find the right bearing in the storeroom. He finds it at 6:30. Installs it in ninety minutes. The WO records sixteen hours of downtime. The actual repair time was under two. Everything else was waiting.*

MTTR is the sum of all the delays between "broken" and "running." A high MTTR means the plant has not prepared for the failures it knows will happen. World-class plants target MTTR under 2 hours on critical assets. Most plants run 6 to 12. The gap is rarely about tech skill. It is about what was staged before the breakdown.

WHY IT MATTERS

Every additional hour of MTTR is lost production and potential cascading effects downstream. High MTTR is almost always a symptom of deeper issues: poor planning for foreseeable failures, inadequate training, insufficient spare parts, missing job plans for known scenarios.

90-DAY ROADMAP

Days 1-3	MTTR analysis by asset and failure mode. Which combinations drive the longest repairs?
Days 3-5	Diagnose delay causes. Parts waiting? Diagnosis difficulty? Access? Specialist needed?
Week 1-2	Improve spare parts staging. Critical spares 24/7 accessible. Emergency repair kits for top 10 scenarios.
Week 2-3	Standardize repair procedures. Job plans for common failures. Step-by-step, tools, LOTO, expected time.
Week 3-6	Cross-train technicians. At least 2 techs per shift capable on each critical asset.
Week 4-6	Repair-vs-replace analysis for chronic failures. Repair cost >50% of replacement in 2 years justifies replacement.

TOP ACTIONS

1. Stage critical spare parts for high-MTTR assets
2. Standardize job plans for common failure modes
3. Cross-train technicians on critical equipment
4. Evaluate repair-vs-replace for chronic failures

BENCHMARK

MTTR: world-class <2h, target <4h, action threshold >8h. Parts-Related Delay: world-class <5%, target <10%, action >20%. Source: SMRP Best Practice 5.4.3, 5.6.1.

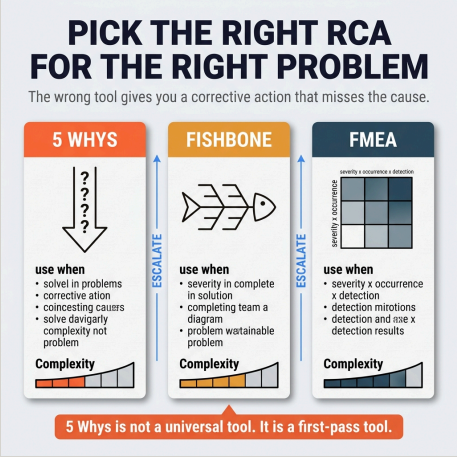
FURTHER READING

- SMRP Best Practice 5.4.3 — Mean Time To Repair
- SAE JA1012 — Guide to the RCM Standard
- ISO 55001:2014 — Asset Management Requirements

RCA Quick Guide

Root cause analysis is not one methodology. It is three, and each has a specific use case. Most plants pick one — usually 5 Whys, because it is the easiest — and apply it to everything. Then they wonder why the corrective actions don't stick.

A simple failure needs a simple tool. A multi-factor failure needs structure. A critical asset needs a risk-ranked proactive approach.



5 WHYS

Best for: Simple, single-cause failures. Linear cause-effect chains. Under a day.

Method: Define the problem. Ask "why?" five times. Verify the root cause with evidence. Define a corrective action that addresses the root, not the symptom.

FISHBONE (ISHIKAWA)

Best for: Multi-factor failures. Team-based. Recurring problems where single-cause assumption failed.

Method: Problem statement at the fish head. Six category branches: Man, Machine, Method, Material, Measurement, Environment. Brainstorm causes. Ask "why" to drill. Rank by likelihood. Verify with data.

FMEA (FAILURE MODE AND EFFECTS ANALYSIS)

Best for: Proactive analysis of critical assets. New equipment. Design or process change review.

Method: List potential failure modes. Rate Severity, Occurrence, Detection (1-10 each). Calculate RPN = S × O × D. Prioritize actions on highest RPN. Re-score after implementation.

WHICH ONE, WHEN

SITUATION	USE
Single failure, non-critical, one day or less	5 Whys
Repeat failure, multi-factor, team-based	Fishbone
Critical asset, proactive, pre-failure	FMEA

You can stack them. A first-pass 5 Whys that does not resolve escalates to Fishbone. A Fishbone that identifies a high-consequence failure mode escalates to FMEA. The ladder goes up when simpler tools do not settle the question.

FURTHER READING

- Taiichi Ohno, *Toyota Production System* — the origin of 5 Whys
- Kaoru Ishikawa, *Guide to Quality Control* — the original Fishbone text
- SAE J1739 — Potential Failure Mode and Effects Analysis
- MIL-STD-1629A — FMEA military standard

The 90-Day Implementation Cadence

The ten playbooks are not a buffet. You cannot execute them all at once. Each consumes planning capacity, technician attention, and supervisor enforcement. Running three simultaneously is hard. Running five will break.

The sequence below assumes you have done the Part 3 self-assessment. Start where your level says to start. Finish a playbook before opening the next one.

LEVEL 1 PLANT (REACTIVE)

Months 1-3: PM Compliance Recovery (PB-PM-001). Get the PM program off the floor. The single highest-leverage play for a Level 1 plant.

Months 2-4: Reactive Work Reduction (PB-RE-001). Run in parallel starting around month 2. Tightly coupled with PM recovery.

Months 4-6: Schedule Compliance Improvement (PB-SC-001). With PMs flowing and reactive shrinking, schedule discipline becomes the next bottleneck.

Months 6-9: Backlog Growth Control (PB-BK-001). By month 6 the backlog shape has shifted. Purge, prioritize, stabilize.

At this point you are operating as a credible Level 2 plant.

LEVEL 2 PLANT (PREVENTIVE)

Months 1-2: Schedule Compliance Improvement (PB-SC-001) if not already running. Level 2 plants almost always have this gap.

Months 2-4: Repeat Failure Investigation (PB-FA-001). Start the RCA discipline on your top 3-5 repeat failures.

Months 4-6: Spare Parts Management (PB-SPM-001). ABC classification and PM kitting.

Months 6-9: Inspection Compliance Recovery (PB-INS-001) if overdue rate is elevated. Regulatory inspections are non-negotiable.

Months 9-12: CBM pilot (PB-CBM-001). Start with 3-5 critical assets. The transition play to Level 3.

LEVEL 3 PLANT (CONDITION-BASED)

Months 1-3: Downtime Reduction (PB-DT-001) if availability below 97%.

Months 2-4: Asset Reliability Improvement (PB-DT-002). MTTR reduction.

Months 4-9: CBM program expansion. From 5 pilot assets to 20-plus critical assets with integrated alerts and trending.

Months 9-12: Repeat Failure elimination at scale. Apply RCA systematically.

A NOTE ON PACING

Each playbook has a ninety-day curve. You are not trying to finish in ninety days. You are trying to bend the curve. A KPI that was declining and is now stable is a win. A KPI that was stable and is now improving is a bigger win. Hold the improvements before reaching for the next playbook.

Expectations high. Excuses low.

References

Organized by type. Start here if you want to go deeper on any topic in this playbook.

SMRP BEST PRACTICES

5.1.1 — Planned Work Percentage

5.1.2 — Reactive Maintenance Work

5.3.1 — Maintenance Backlog

5.3.2 — Work Order Planning and Completion

5.3.3 — WO Generation and Identification

5.3.4 — Break-in Work Percentage

5.3.5 — Schedule Compliance

5.3.7 — Proactive Work Percentage

5.4.1 — Equipment Reliability

5.4.2 — Overall Equipment Effectiveness

5.4.3 — Mean Time To Repair

5.5.2 — PdM Program Coverage

5.5.6 — PM/PdM WO Completion Rate

5.6.1 — MRO Inventory Management

ISO STANDARDS

ISO 55000:2014 — Asset Management Overview

ISO 55001:2014 — System Requirements

ISO 55002:2018 — Application Guidelines

ISO 14224:2016 — Reliability Data

ISO 17359:2018 — Condition Monitoring

ISO 10816 — Mechanical Vibration Evaluation

SAE STANDARDS

SAE JA1011 — RCM Evaluation Criteria

SAE JA1012 — Guide to the RCM Standard

SAE J1739 — FMEA in Design and Manufacturing

INSPECTION CODES (REGULATORY)

API 510 — Pressure Vessel Inspection

API 570 — Piping Inspection

API 653 — Tank Inspection

OTHER STANDARDS

MIL-STD-1629A — FMEA military standard

OSHA 1910.147 — LOTO

ANSI Z244.1 — Lockout/Tagout

NETA MTS — Maintenance Testing Specs

BOOKS

Doc Palmer, *Maintenance Planning and Scheduling Handbook* (3rd ed.)

Ron Moore, *Making Common Sense Common Practice* (4th ed.)

John Moubray, *Reliability-Centered Maintenance II*

Terry Wireman, *Benchmarking Best Practices* (3rd ed.)

Kaoru Ishikawa, *Guide to Quality Control*

Taiichi Ohno, *Toyota Production System*

Paul Barringer, *Weibull Analysis*

ORGANIZATIONS

SMRP — smrp.org

Reliabilityweb / Uptime Magazine

IMC — International Maintenance Conference

Maintenance World journal

A FINAL NOTE ON SOURCES

This playbook does not invent anything. Every benchmark, every methodology, every threshold has a citation. If you disagree with a number, disagree with the source. The job of the maintenance manager is not to argue the target. It is to meet it.

About + Keep Going

WHO WROTE THIS

My name is Ivan Getov. I help plants and manufacturing teams perform better by fixing the system, not blaming the people.

I grew up between two cultures. Born in Bulgaria, came up professionally in the U.S. Each taught me something the other didn't. Bulgarian directness doesn't soften a hard truth. It names it. American workplaces, on the other hand, will go to great lengths to avoid a hard conversation. Both modes fail if you run them pure. Direct without context sounds like aggression. Conflict-avoidance without accountability becomes a plant where nobody says what is actually broken. What I have tried to carry forward is the blend. Direct about the system. Respectful toward the people inside it.

I have spent most of my working life inside industrial operations, from the floor up. What I learned along the way is simple and ought to be obvious: most broken plants are not broken because the people running them are lazy or incompetent. They are broken because the system underneath them is designed badly, held loosely, or abandoned entirely after the last consultant left.

This playbook is drawn from that work. From the standards and teachers who have shaped modern maintenance. From the maintenance managers and plant supervisors I have watched turn plants around. And from the quiet, unglamorous truth that you cannot improve what you are not willing to measure, and you cannot measure what you do not care enough to enforce.

KEEP GOING

If this was useful, there is more where it came from.

BLOG

getov.xyz — long-form pieces on leadership, maintenance, and the gap between what companies say and what actually happens.

LINKEDIN

linkedin.com/in/ivan-getov-mba — shorter-form content on the same themes, three times a week.

SUBSTACK

getov.substack.com — longer newsletter pieces, roughly weekly. Interviews, stories, and the unedited version of ideas that show up on the blog.

ONE LAST THING

Fix the system. Back the people running it. Hold the standard when it is inconvenient, because that is when it counts. The rest is commentary.

Expectations high.
Excuses low.

— Ivan